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Goal-derived information constraints for dual control

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Abstract

This paper addresses dual control, i.e., the design of a control system that simultaneously must collect data for system identification and regulate the system towards a target. We treat dual control as a Bayesian inference problem and propose a method that favours identification when standard inputs are predicted to lead to insufficient information. We formulate a Bethe free energy functional over the factor graph of a planning model and impose an information gain constraint between future outputs and the beliefs over unknown parameters and hidden states, given the system inputs. This functional is minimized through message passing, producing a dual controller that does not require separate tuning of a probing weight. Our approach is tested on a benchmark integrator and a quadcopter navigation problem, both with unknown parameters. The constrained Bethe free energy controller performs on par with optimally tuned state-of-the-art dual controllers, without tuning.

Keywords: Dual control, Bethe free energy, factor graphs, message passing, information gain

1. Introduction

Dual control refers to the problem of driving a system with unknown parameters, where one system must simultaneously perform online system identification and adaptive control (Feldbaum, 1960). Most dual controllers consist of a model predictive control (MPC) cost with an added excitation or uncertainty term (Mesbah, 2018; Heirung et al., 2017). This necessitates balancing probing and regulation, also known as the exploration versus exploitation trade-off (Levine, 2018). Dual controllers typically tune a weight term to a specific system and task, but that leaves the balance frozen (Larsson et al., 2013; Mesbah, 2018). The aim is to design a dual controller that re-evaluates the need for information gathering every time step.

We approach dual control as a Bayesian inference problem, implemented as message passing on a factor graph (Loeliger et al., 2007). This is an efficient inference scheme as it leverages the factorization structure of dynamical models and flexible as it allows for approximations and constraints specific to individual nodes and edges (Levine, 2018; Hoffmann and Rostalski, 2017). The Bethe free energy is a variational objective over the local beliefs of a factor graph, and is a general framework encompassing a number of well-known inference algorithms, such as belief propagation and variational message passing (Yedidia et al., 2005; Şenöz et al., 2021).

MPC with integrated experiment design (MPC-X) imposes an information requirement on controls, but requires independent weight tuning (Larsson et al., 2013). We formulate dual control as constrained Bethe free energy minimization on a factor graph, with information constraints whose floors come from the need to reach a goal with sufficient certainty (Definition 3). Their KKT multipliers act as probing weights that vanish once the controls are sufficiently informative (Theorem 5, Lemma 6). Since the parameter belief stays fixed within the plan, the controller is an explicit dual controller in the sense of Mesbah (2018). We test our approach on a benchmark integrator with unknown gain and a quadcopter navigation task with an unknown compass fault, demonstrating competitive performance without probing weight tuning (Section 5).

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2. Problem statement

Consider discrete-time systems with unknown parameters,

$$x_k = f(x_{k-1}, u_k, \theta) + w_k\,, \qquad y_k = g(x_k) + v_k\,, \tag{1}$$

where $x_k \in \mathbb{R}^{n_x}$ is the state, $u_k \in \mathcal{U} = [-1, 1]^{n_u}$ the input with the actuator range scaled to $\mathcal{U}$, $y_k \in \mathbb{R}^{n_y}$ the output, and $\theta \in \mathbb{R}^{n_\theta}$ a vector of parameters. The process noise w_k and the measurement noise v_k are white. The controller knows the model structure, i.e., the functions f and g and the noise families, and a prior distribution over θ, but not θ itself. At time t, the controller has the data $y_{1:t}, u_{1:t}$ and selects the next input u_{t+1}. Its task is to regulate the output to a target $y_* \in \mathbb{R}^{n_y}$ within a tolerance $S_* \succ 0$ but make accurate predictions as well. So we evaluate controllers by their prediction error and their goal cost incurred over a trial from time 1 to T,

$$G = \frac{1}{T}\sum_{k=1}^{T} \frac{1}{2}\|y_k - \mu_k\|^2_{\Sigma_k^{-1}} + \frac{1}{2}\|y_k - y_*\|^2_{S_*^{-1}}\,, \tag{2}$$

where μ_k and Σ_k are the mean and covariance matrix of the predictive distribution. How should a receding-horizon controller gather information about θ such that it minimizes G?

3. Probabilistic models

We frame dual control as a Bayesian inference problem. Let $\tau = (t+1, \ldots, t+K)$ be a finite horizon and $\phi_* = (y_*, S_*)$. A probabilistic state-space model over the planning horizon is:

$$p(y_\tau, x_\tau, u_\tau, \theta, x_t \,|\, \phi_*) \propto \underbrace{p(x_t)p(\theta)}_{\text{priors}} \prod_{k\in\tau} \underbrace{p(y_k \,|\, x_k)}_{\text{likelihood}} \underbrace{p(x_k \,|\, x_{k-1}, \theta, u_k)}_{\text{state transition}} \underbrace{p(u_k)}_{\text{control prior}} \underbrace{p(y_k \,|\, \phi_*)}_{\text{goal prior}}\,. \tag{3}$$

With $s = (y_\tau, u_\tau, \theta, x_\tau, x_t)$, the model is a product of factors, $p(s) \propto \prod_{a\in\mathcal{V}} f_a(s_a)$ where every factor f_a of the set $\mathcal{V}$ depends on a subset s_a of the variables. A Forney-style factor graph (FFG) draws every factor as a node and every variable as an edge between at most two nodes, and an equality node joins the edges of a variable with more than two factors (Loeliger et al., 2007). Unlike bipartite factor graphs, FFGs make sequential computations explicit. Fig. 1 shows the FFG of (3) over the horizon. Since future outputs are not measured, "goal prior" factors terminate the output edges and bias the inference towards trajectories, and thereby towards inputs, that realize the goal. We encode the target of Section 2 as the Gaussian goal prior $p(y \,|\, \phi_*) = \mathcal{N}(y \,|\, y_*, S_*)$.

The controller infers the belief of the next input u_{t+1} by message passing, with a forward sweep to the end of the horizon and a backward sweep in return (Section 4.3). The two messages on an edge multiply to its belief, as Fig. 1 shows for u_{t+1}. However, exact inference in (3) does not yield a dual controller. It treats the goal priors as observations, so it moves the parameter belief towards parameters under which the goal is easy to reach, an issue known as "optimism" in control as inference (Levine, 2018). Moreover, the exact posterior does not reward inputs that reveal θ. Section 4 constrains the inference towards dual control.

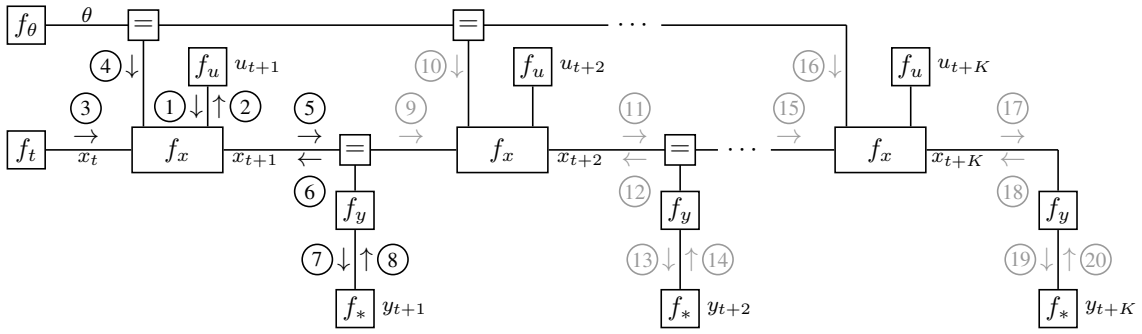


Figure 1: FFG of the planning model (3), whose nodes are $f_t = p(x_t)$, $f_\theta = p(\theta)$, $f_u = p(u_k)$, $f_x = p(x_k \,|\, x_{k-1}, \theta, u_k)$, $f_y = p(y_k \,|\, x_k)$ and $f_* = p(y_k \,|\, \phi_*)$. Equality nodes (=) enforce consistency (Loeliger et al., 2007). On the edge u_{t+1}, the node f_u sends ① $= p(u_{t+1})$, and f_x sends ② $= \nu(u_{t+1})$, computed from messages on its other edges.

4. Inference

4.1. Bethe free energy and constraints

The Bethe free energy distributes a free energy functional into average energy terms associated with nodes and entropy terms associated with edges (Yedidia et al., 2005; Şenöz et al., 2021). For the nodes $\mathcal{V}$, the edges $\mathcal{E}$ between them and a variational model $q(s)$, it reads

$$\mathcal{B}[q] = \sum_{a \in \mathcal{V}} \int q_a(s_a) \ln \frac{q_a(s_a)}{f_a(s_a)} \, \mathrm{d}s_a + \sum_{i \in \mathcal{E}} \mathbb{H}[q_i] \,, \tag{4}$$

where the beliefs q_a and q_i are the distributions of the node a and the edge i. Normalization constraints, $\int q_a(s_a) \, \mathrm{d}s_a = 1$, ensure the beliefs are valid probability distributions. Marginalization constraints, $\int q_a(s_a) \, \mathrm{d}s_{a \setminus i} = q_i(s_i)$ for every node a and every edge i of a, make them consistent (Şenöz et al., 2021). Under these local constraints, the stationary points of $\mathcal{B}$ are the fixed points of belief propagation (Heskes, 2002). Additional constraints yield other message passing algorithms (Yedidia et al., 2005). We employ factorization and form constraints.

Definition 1 (Factorization constraints) *Let* $\mathrm{in}(a)$ *be the set of input edges of node* a*. The belief* q_a *equals the factor* f_a *times the edge beliefs* q_i *on its input edges,*

$$q_a(s_a) = f_a(s_a) \prod_{i \in \mathrm{in}(a)} q_i(s_i) \,. \tag{5}$$

Definition 2 (Form constraints) *The belief* q_b *is pinned to the form of the factor* f_b*,*

$$q_b(s_b) = f_b(s_b) \quad \textit{for some } b \in \mathcal{V} \,. \tag{6}$$

We assume a separate Bayesian filtering procedure produces posterior distributions for the current state x_t and the parameters θ. In the planning model, the marginal beliefs over these are form constrained to the posteriors $q(\theta) = p(\theta \,|\, y_{1:t}, u_{1:t})$ and $q(x_t) = p(x_t \,|\, y_{1:t}, u_{1:t})$, so that the goal priors cannot cause optimism. The transition and measurement functions f and g are known, so the factorization constraints (5) keep their factors in the node beliefs.

We propose to add information gain constraints. Recall that the information gain between the variables a_1 and a_2 given a third variable a_3 under a belief q_a is (Cover and Thomas, 1999)

$$\mathbb{I}_a[a_1 : a_2 \,|\, a_3] = \int q_a(a_1, a_2 \,|\, a_3) \ln \frac{q_a(a_1, a_2 \,|\, a_3)}{q(a_1 \,|\, a_3)\, q(a_2 \,|\, a_3)} \mathrm{d}(a_1, a_2)\,. \tag{7}$$

For the state-space model, we constrain the information $\mathbb{I}_{t+1}[y_{t+1} : x_{t+1}, \theta \,|\, u_{t+1}]$ that the next output carries about the unknown state and parameters, given the next input. It follows from the joint belief of the measurement and transition nodes,

$$q(y_{t+1}, x_{t+1}, \theta \,|\, u_{t+1}) = q(y_{t+1} \,|\, x_{t+1}) \int q(x_{t+1} \,|\, x_t, \theta, u_{t+1})\, q(\theta)\, q(x_t)\, \mathrm{d}x_t\,. \tag{8}$$

Definition 3 (Information gain constraints) *The information gain between two variables in belief q_a given a third has to exceed a lower bound β:*

$$\mathbb{I}_a[a_1 : a_2 \,|\, a_3] \geq \beta\,. \tag{9}$$

The constraint biases the control posterior $q(u_{t+1})$ towards informative inputs, and β, the information floor, sets the strength of this bias. A low floor has no effect, whereas a high floor overrides the goal, so we derive the floor from the goal. Let $\mu(u)$ and $\Sigma(u)$ be the mean and covariance of the predictive distribution $q(y_{t+1} \,|\, u_{t+1} = u)$. The covariance matters only at controls that reach the goal, so we weight every control by its prior probability and by the goal likelihood of its mean prediction,

$$\tilde{p}_t(u) \propto p(u)\, \mathcal{N}\big(\mu(u) \,|\, y_*, S_*\big)\,, \qquad \bar{\Sigma}_{t+1} = \mathbb{E}_{\tilde{p}_t}\big[\Sigma(u)\big]\,. \tag{10}$$

Information is thus needed when the covariance $\bar{\Sigma}_{t+1}$ of the goal-reaching inputs exceeds the tolerance. Since no input removes the noise on the output, whose covariance R is known or estimated and enters $\bar{\Sigma}_{t+1}$, we add R to the tolerance. The floor is then the excess entropy

$$\beta_t = \mathbb{H}\big[\mathcal{N}(0, \bar{\Sigma}_{t+1})\big] - \mathbb{H}\big[\mathcal{N}(0, R + S_*)\big] = \tfrac{1}{2} \ln \big|\bar{\Sigma}_{t+1}(R + S_*)^{-1}\big|\,. \tag{11}$$

Predictions within tolerance, $\bar{\Sigma}_{t+1} \preceq R + S_*$, give $\beta_t \leq 0$ and leave the constraint inactive, whereas the further the predictions exceed the tolerance, the more information the constraint demands.

4.2. Stationary control posterior

The factorization and form constraints restrict the family of beliefs, so we substitute them into the Bethe free energy before we add multipliers for the remaining constraints (Şenöz et al., 2021).

Lemma 4 (Bethe collapse) *The Bethe free energy (4) of the model (3) reduces, under normalization, marginalization, factorization and form constraints, to*

$$\mathcal{B}[q] = \sum_{k \in \tau} D_{KL}\big[q(u_k) \,\|\, p(u_k)\big] - \mathbb{E}_{q(y_k)}\big[\ln p(y_k \,|\, \phi_*)\big]. \tag{12}$$

The proof is in Appendix A. Since (12) agrees with (4) on the constraint set, the two share their constrained stationary points. The form constraints fix $q(x_t)$ and $q(\theta)$, and the node beliefs (5) meet the marginalization at their input edges, because the transition and measurement factors are normalized

conditional densities. Only the normalization of the control beliefs and the marginalization at the outputs of the transition and measurement nodes remain. With multipliers ψ_k, λ_{y_k} and λ_{x_k} for these constraints and $\gamma \geq 0$ for the information gain constraint, the constrained Bethe Lagrangian is

$$\begin{aligned}\mathcal{L}[q] = \mathcal{B}[q] + \gamma\Big(\beta_t - \mathbb{I}_{t+1}\big[y_{t+1} : x_{t+1}, \theta \,|\, u_{t+1}\big]\Big) \\ + \sum_{k\in\tau}\Big(\psi_k\big(\int q(u_k)\,\mathrm{d}u_k - 1\big) + \int \lambda_{y_k}(y_k)\big(q(y_k) - \int p(y_k \,|\, x_k)\, q(x_k)\,\mathrm{d}x_k\big)\mathrm{d}y_k \\ + \int \lambda_{x_k}(x_k)\big(q(x_k) - \int p(x_k \,|\, x_{k-1}, \theta, u_k) q(x_{k-1})\, q(\theta)\, q(u_k)\,\mathrm{d}(x_{k-1}, \theta, u_k)\big)\mathrm{d}x_k\Big).\end{aligned} \tag{13}$$

This Lagrangian is to be minimized with respect to the marginal beliefs of each future action.

Theorem 5 (Control posterior) *At the stationary point of* (13)*, every input u_k has the belief*

$$q^\star(u_k) \propto p(u_k) \exp\Big(\sum_{m=k}^{t+K} \mathbb{E}_{q(y_m \,|\, u_k)}\big[\ln p(y_m \,|\, \phi_*)\big] + \gamma_k\, \mathbb{I}_k(u_k)\Big), \tag{14}$$

where $q(y_m \,|\, u_k)$ is the prediction of y_m that the marginalization constraints propagate from the input value u_k and the beliefs of the other inputs. The information gain $\mathbb{I}_{t+1}(u_{t+1})$ at the input value u_{t+1} has the weight $\gamma_{t+1} = \gamma$, and $\gamma_k = 0$ for later inputs.

The proof is in Appendix A. In words, the approximate posterior is the control prior times a pseudo-likelihood, consisting of the exponentiated cross-entropy between predictions and goals, and the potential information gained on states and parameters from predicted future observations. The cross-entropy term for a Gaussian goal prior and a prediction of y_k with mean μ_k and covariance Σ_k is $\mathbb{E}[\ln p(y_k \,|\, \phi_*)] = -\frac{1}{2}\|\mu_k - y_*\|^2_{S_*^{-1}} - \frac{1}{2}\mathrm{tr}\big(S_*^{-1}\Sigma_k\big) + \text{constants}$. These two terms are the tracking and caution terms of dual control (Bar-Shalom and Tse, 1974). For $\gamma = 0$, the mode of $q^\star$ is a cautious controller. A fixed weight on the information gain of every step gives an explicit dual MPC. With weight one, the stationary points coincide with those of expected free energy minimization (Kouw, 2026). Instead of tuning, we solve for γ by meeting the KKT conditions of the information gain constraint. For the posterior (14), they read

$$\gamma \geq 0\,, \quad \mathbb{I}_{t+1}[q^\star] \geq \beta_t\,, \quad \gamma\big(\beta_t - \mathbb{I}_{t+1}[q^\star]\big) = 0\,, \tag{15}$$

which are dual feasibility, primal feasibility and complementary slackness. The joint belief (8) contains only factors and fixed beliefs, so at the posterior $q^\star(u_{t+1}; \gamma)$ for the multiplier γ, the constraint takes the value

$$h(\gamma) = \mathbb{I}_{t+1}[q^\star] = \mathbb{E}_{q^\star(u_{t+1};\gamma)}\big[\mathbb{I}_{t+1}(u_{t+1})\big]\,. \tag{16}$$

A larger multiplier tilts the posterior towards informative inputs, so h grows with γ. However, h cannot exceed the information gain of the most informative inputs that the posterior can select. The posterior contains the prior factor $p(u_{t+1})$ and thus ignores inputs of zero prior probability. The bound is the essential supremum of the information gain under the prior,

$$\mathbb{I}^{\max} = \operatorname*{ess\,sup}_{u\sim p} \mathbb{I}_{t+1}(u) = \min\Big\{c : \Pr_{u\sim p}\big[\mathbb{I}_{t+1}(u) > c\big] = 0\Big\}\,, \tag{17}$$

the smallest value that the information gain exceeds only with prior probability zero. For a continuous information gain and a prior with support $\mathcal{U}$, it is the maximum of $\mathbb{I}_{t+1}$ over $\mathcal{U}$.

Lemma 6 (Information multiplier) *Fix all beliefs except $q(u_{t+1})$, and let $\mathbb{I}_{t+1}(u)$ be bounded on $\mathcal{U}$ and not constant on any set of inputs with prior probability one. Then h is continuous and strictly increasing, and $h(\gamma) \to \mathbb{I}^{\max}$ as $\gamma \to \infty$. Hence the KKT conditions (15) have the unique solution*

$$\gamma = 0 \quad \text{if } \beta_t \leq h(0)\,, \qquad \gamma = h^{-1}(\beta_t) \quad \text{if } h(0) < \beta_t < \mathbb{I}^{\max}\,, \tag{18}$$

and no solution if $\beta_t \geq \mathbb{I}^{\max}$.

The proof is in Appendix A. If the posterior for $\gamma = 0$ already meets the floor, the controller does not probe. Otherwise, the multiplier is the probing weight at which the posterior meets the floor exactly, so a higher floor demands a larger weight.

4.3. Message passing

Message passing computes the posterior (14) by sweeps over the FFG of Fig. 1, each with a forward pass that predicts the outputs and a backward pass that scores them against the goal. From the form-constrained beliefs ③ $= q(x_t)$ and ④ $= q(\theta)$, every conditional node sends the marginal of its belief (5) forward, such as the predicted state ⑤ and output ⑦, as in the prediction steps of the Kalman filter (Loeliger et al., 2007). The backward pass starts from the goal priors, such as ⑧ $= p(y_{t+1} \,|\, \phi_*)$, and every conditional node c sends to its input edge j the exponentiated expectation of the log-messages on its outputs,

$$\nu_{c \to j}(s_j) = \exp\Big(\mathbb{E}_{q_c(s_c \,|\, s_j)}\Big[\sum_{l \in \mathrm{out}(c)} \ln \nu_{d_l \to l}(s_l)\Big]\Big), \tag{19}$$

where $q_c(s_c \,|\, s_j)$ is the belief (5) with s_j fixed and d_l is the node that consumes the output l. The equality node of x_{t+1} multiplies the messages from the measurement node and the next transition node into ⑥, from which the transition node computes ②. Averaging log-messages, instead of messages as in belief propagation, gives the backup of control as inference with fixed dynamics, which avoids optimism (Levine, 2018). For linear Gaussian nodes, every backward message carries a quadratic cost-to-go, as in LQ control on factor graphs (Hoffmann and Rostalski, 2017).

Since every slice repeats the nodes of the first, the messages ⑨–⑭ at $t+2$ and ⑮–⑳ at $t+K$, drawn lighter in Fig. 1, have the functional forms of ③–⑧, in order, except that ⑱ lacks a message from a next transition node. The backward pass updates every control belief to the product of the messages on its edge, such as ① and ②, and the constraint adds $e^{\gamma\, \mathbb{I}_{t+1}(u)}$ on the edge of u_{t+1}. For a fixed multiplier, every update minimizes $\mathcal{L}$ over one control belief, so the sweeps decrease $\mathcal{L}$ monotonically. The planner linearizes the transition nodes at their incoming means and keeps every control belief at its mode. As these coordinate updates converge linearly, the experiments find the same fixed point by quasi-Newton steps (Section S2.3 of the supplementary material).

5. Experiments

We perform experiments on systems with unknown parameters[1]. To control for model flexibility, all controllers in a comparison share the model and the filter, and the predictive controllers have the same horizon length K.

1. Code: https://github.com/biaslab/L4DC2027-InfoConstraints

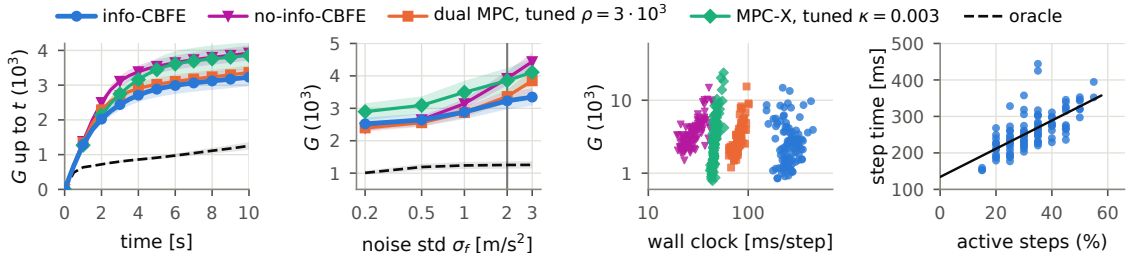


Figure 2: At $\sigma_f = 2$ m/s^2, the untuned info-CBFE costs as little as the dual MPC tuned at this noise and less than no-info-CBFE and MPC-X. Ribbons show one standard error over 99 flights, and the dashed line shows an oracle that knows the wind and the fault. (a) Performance (2) accumulated over the flight at $\sigma_f = 2$ m/s^2. (b) Performance against IMU noise (vertical line indicates σ_f for (a) and (c)). (c) Performance against per-step wall-clock time. (d) Per-step wall-clock time of info-CBFE versus the proportion of constraint activations, with a least-squares regression (black).

Baselines The proposed controller is called *info-CBFE*, and the ablation without the information gain constraint is called *no-info-CBFE*. As state-of-the-art methods, we compare to *dual MPC*, which minimizes $-\ln q^\star(u_k)$ for each $k \in \tau$ with the fixed weight $\gamma_k = \rho/2$. It comprises the goal tracking, caution and probing terms of explicit dual control (Mesbah, 2018; Heirung et al., 2017), and differs from info-CBFE only in how it weighs the information. *MPC-X* minimizes a nominal goal tracking cost subject to the experiment-design constraint of Larsson et al. (2013). This constraint demands enough information that, with probability 0.95, the estimation error raises the goal cost by at most $1/\kappa$, so the required information grows with the accuracy κ. We tune ρ and κ on held-out seeds at the nominal sensor noise.

5.1. Integrator with unknown gain

The benchmark of Åström and Helmersson (1986) consists of $x_k = x_{k-1} + b\,u_k + w_k$ and $y_k = x_k$ that has to be driven to zero over $T = 50$ steps from $x_0 = 0$. The noise is zero-mean Gaussian distributed, $w_k \sim \mathcal{N}(0, \sigma^2)$, and the gain is unknown, $\theta = b \sim \mathcal{N}(b_0, 1)$. The parameter belief is the Kalman filter $\mathcal{N}(\hat{b}_t, P_t)$, so the gain adds the variance $P_t u^2$ to the prediction. The information of the transition node is $\frac{1}{2}\ln(1 + \ell_t(u))$ with $\ell_t(u) = P_t u^2/\sigma^2$. With the goal prior $\mathcal{N}(0, \sigma^2)$, the floor is $\beta_t = \frac{1}{2}\ln\left((\sigma^2 + P_t\,\mathbb{E}_{\tilde{p}_t}[u^2])/(2\sigma^2)\right)$. We set $\sigma = 0.1$ and run 200 episodes for each prior mean $b_0 \in \{0, 0.5, 1, 2, 4\}$, with b drawn from its prior. Tuning on 50 held-out episodes per prior mean selects $\rho = 5$ from $\{0, 0.5, 1, 2, 5, 10, 20, 50\}$ and $\kappa = 0.1$ for MPC-X from $\{0.01, 0.03, 0.1, 0.3, 1\}$. The optimal dual controller follows from dynamic programming on a 121×121 grid without an input bound. All other controllers keep $u_k \in \mathcal{U} = [-1, 1]$ (Section S2.6 of the supplementary material).

Results Without tuning, info-CBFE comes within 12% of the dual MPC with the best probing weight at every prior mean. Over all prior means, it attains $G = 1.41$, against 1.34 for the dual MPC with the tuned $\rho = 5$, 1.35 for MPC-X and 1.04 for the optimal dual controller, which has no input bound. The paired differences to the dual MPC and MPC-X are 0.069 [0.005, 0.155] and 0.058 [−0.020, 0.158]. Outside $5 \le \rho \le 10$, the dual MPC exceeds the G of info-CBFE by at least 17% at some prior mean (Fig. 3 left). With $\rho \le 2$, it applies no input under $b_0 = 0$, where its G is 12.8, and $\rho = 50$ over-probes, which raises its G by 22% over all prior means.

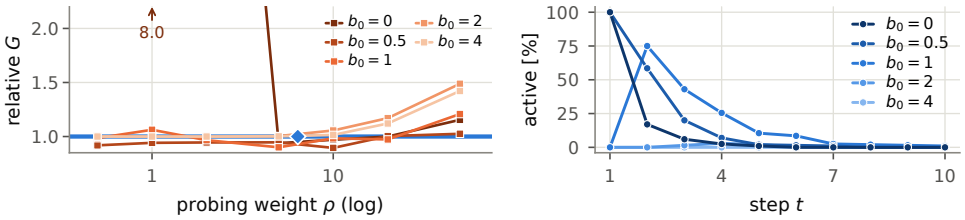


Figure 3: (Left) G of the dual MPC relative to that of info-CBFE against its probing weight ρ, per prior mean b_0 over 200 episodes. The blue line marks info-CBFE, which needs no tuning, and the diamond marks twice the median of its active multipliers. (Right) Proportion of episodes, per prior mean b_0, in which the information constraint of info-CBFE is active. The constraint acts only in the first steps.

The multiplier probes only while the gain is too uncertain for the goal (Fig. 3 right). Its constraint is active on 2.0% of the steps, at the first step of every episode for $b_0 \leq 0.5$ and never for $b_0 = 4$. The active multipliers have a median of 3.2, which corresponds to $\rho = 6.4$, inside the window of good weights. Without the constraint, no-info-CBFE applies no input under $b_0 = 0$, because every input only spreads the prediction while the sign of the gain is unknown.

5.2. Quadcopter navigation under a faulty compass heading

Consider a quadcopter with state $x_k = (p_k, \dot{p}_k, f_k)$, for position, velocity and force, and control space $\mathcal{U} = [-1, 1]^3$ that has to navigate to a target 5 m away and 1 m higher within $S_* = (0.03\,\mathrm{m})^2 I$. The trial is $T = 20$ steps with step size $\Delta = 0.5$ s, and the controllers run a horizon of length $K = 6$. Its equations of motion are (Mahony et al., 2012),

$$
\begin{aligned}
f_k &= \alpha_k \begin{bmatrix} \Gamma\, n_k^{\mathrm{h}} \\ \eta\, n_k^{\mathrm{v}} \end{bmatrix} + \alpha_k \Pi_k \big(c_k - d\,\dot{p}_{k-1}\big) + w_k\,, \qquad \Gamma = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}, \\
\dot{p}_k &= \dot{p}_{k-1} + \Delta\,(f_k - g e_3)\,, \qquad p_k = p_{k-1} + \Delta\,\dot{p}_{k-1} + \tfrac{1}{2}\Delta^2 (f_k - g e_3)\,,
\end{aligned}
\tag{20}
$$

where $\alpha_k = g\,(1 + u_{k,1}/2)$ is the thrust per mass, $n_k = (\cos\varphi \sin\vartheta, -\sin\varphi, \cos\varphi\cos\vartheta)$ the thrust axis for the roll $\varphi = 0.4\,u_{k,2}$ and the pitch $\vartheta = 0.4\,u_{k,3}$, n_k^{h} and n_k^{v} its horizontal and vertical parts, and $\Pi_k = I - n_k n_k^\top$ the projection onto the rotor plane. A position fix and an IMU measure $y_k = (p_k + v_k^{\mathrm{p}}, f_k + v_k^{\mathrm{f}})$ with $v_k^{\mathrm{p}} \sim \mathcal{N}(0, 0.3^2 I)$ and $v_k^{\mathrm{f}} \sim \mathcal{N}(0, \sigma_f^2 I)$, and $w_k \sim \mathcal{N}(0, 0.2^2 I)$ collects unmodelled forces.

The controller learns the parameters $\theta = (c_k, d, a, b, \eta)$, which all enter (20) linearly. Magnetic interference rotates its heading by an unknown angle $\chi \sim \mathcal{U}[-2.6, 2.6]$ rad so that $(a, b) = (\cos\chi, \sin\chi)$ (Kok and Schön, 2016). Rotor drag grows with the thrust and with the airspeed in the rotor plane (Martin and Salaün, 2010; Kai et al., 2017; Faessler et al., 2017), so d scales the drag of its own velocity and $c_k = d\,\omega_k$ the drag of the wind ω_k, whose Weibull-distributed mean speed (Justus et al., 1978) and direction fluctuate with correlation times of 1 to 10 s. The prior on (a, b) matches the moments of the fault distribution. An extended Kalman filter tracks x_k and θ, and c_k and the gains are modelled as random walks. When computing the information floor β_t, we average over 4096 inputs drawn uniformly from $\mathcal{U}$.

Results From 1.5 s on, info-CBFE has accumulated the least cost, and it costs less than MPC-X at every IMU noise (Fig. 2(a) and (b)). Since the seeds 1000–1099 informed the design, we report the fresh seeds 4000–4099 (Section S2.5 of the supplementary material). We drop seed 4076, on which even an oracle that knows the wind and the fault ends farther from the target than it started. At $\sigma_f = 2$ m/s^2, info-CBFE attains $G = 3230$, against 3363 for the dual MPC, 3857 for MPC-X and 3922 for no-info-CBFE, with paired gaps of 133 [-312, 582], 626 [332, 1000] and 692 [195, 1184]. Info-CBFE thus matches the tuned dual MPC at every noise and cuts the learning cost, the cost above that of the oracle, by 24% against MPC-X and 26% against no-info-CBFE. No-info-CBFE ties with info-CBFE only up to $\sigma_f = 1$ m/s^2. The constraint turns the caution of no-info-CBFE into exploration. While the rotation is unknown, every tilt spreads the predicted position, so the trace term holds the first tilt of no-info-CBFE at 7°. The constraint is active at the first step of every flight and the second of 96%. It raises the first tilt of info-CBFE to 30°, so that the first measurements reveal the rotation, and acts again on 26% of the later steps. No-info-CBFE therefore accrues 674 [266, 1087] of its gap to info-CBFE within the first 4 s (Fig. 2(a)). MPC-X, in contrast, barely probes and produces occasional bad flights. Its requirement changes the plan on only 0.05% of the test steps, so it tilts by 31° from the first step and acts on its point estimate while noisy measurements correct it. Its median cost lies 13% above that of info-CBFE, whereas its worst 10% of flights cost 36% more. These flights keep adding cost after info-CBFE levels off, so the gap of MPC-X grows from 449 [264, 663] at 4 s to 626 at 10 s. Owing to the multiplier search, info-CBFE takes a median of 255 ms per step, against 28 to 80 ms for the other controllers, and 4 ms more per percent of active steps (Fig. 2(c) and (d)). An active step takes a median of 0.62 s, longer than the control period of 0.5 s.

6. Discussion

The information (9) has a closed form when the parameters enter the mean linearly with Gaussian noise, whereas nonlinear parameterizations require an approximation. Whether the floor can be attained depends on a parameter-induced covariance that is proportional to the noise covariance.

Constraining only the next node forces the information to come from one measurement, which results in greedy identification plans even though the regulation plans are non-greedy. Constraints on later nodes would spread the probing over the plan, but they require a belief that learns within the plan, as in implicit dual control (Mesbah, 2018). In other words, the belief $q(\theta)$ must not be form constrained, but allowed to change over the course of the plan.

Closed-loop stability remains an open question as well as the consistency of the parameter estimates after the constraint switches off. Future work could combine information gain constraints with chance constraints to plan for avoiding parts of state or control space (van de Laar et al., 2021).

7. Conclusion

We showed that a goal-derived information constraint turns a message passing planner on a factor graph into a dual controller. Variational calculus yields the posterior of the next input as the product of the control prior message and a message that weighs the expected goal log-density against the information gain. The KKT multiplier of the constraint switches on when the entropy of the goal-weighted prediction exceeds that of the goal tolerance plus the output noise. On a quadcopter with an unknown compass fault, the controller matched a dual MPC tuned at the nominal noise and cut

the learning cost by 24% against MPC-X. Future work concerns stability and consistency analysis as well as further constraints.

Appendix A. Proofs

Proof of Lemma 4 All instances of q_a being equal to f_a are cancelled from (4). The form constraints pin q_t to f_t and q_θ to f_θ, while the factorization constraints cancel the transition and measurement factors. The equality node beliefs ensure that the edge beliefs of the states and parameters are counted only once (4). The only terms that remain are the divergences of the control beliefs from their priors and the expected goal log-densities. Section S1.1 of the supplementary material writes out every step. ■

Proof of Theorem 5 The proof sets the functional derivatives of the Lagrangian (13) to zero. The form constraints fix $q(x_t)$ and $q(\theta)$, so only the beliefs of the outputs, states and inputs vary, and we treat them in the order of the backward pass. An output belief $q(y_k)$ enters only the goal term of (12) and its own marginalization constraint, so its multiplier is the goal log-density, $\lambda_{y_k} = \ln p(y_k \mid \phi_*)$. A state belief $q(x_k)$ enters its own constraint and the constraints of the measurement and next transition nodes, which consume x_k, so its multiplier collects theirs,

$$\lambda_{x_k}(x_k) = \mathbb{E}_{p(y_k \mid x_k)}\left[\lambda_{y_k}(y_k)\right] + \mathbb{E}_{q_{k+1}(s_{k+1} \mid x_k)}\left[\lambda_{x_{k+1}}(x_{k+1})\right], \tag{21}$$

where $q_{k+1}(s_{k+1} \mid x_k) = p(x_{k+1} \mid x_k, \theta, u_{k+1})\, q(\theta)\, q(u_{k+1})$ is the next transition belief at a fixed x_k, and the second term vanishes for $k = t + K$. With the message $\nu_{d_l \to l} = \exp(\lambda_l)$, this recursion is the backward pass (19). Unrolling it turns $\lambda_{x_k}(x_k)$ into the summed expected goal log-densities of the outputs y_τ given x_k. Finally, an input belief $q(u_k)$ enters its divergence from the prior, its normalization, the constraint of the state x_k that it drives and, for $k = t + 1$, the information gain, which is linear in $q(u_{t+1})$. Setting its derivative to zero gives the Gibbs form (14) of control as inference (Levine, 2018; Wainwright et al., 2008). Section S1.2 of the supplementary material writes out every derivative. ■

Proof of Lemma 6 With all other beliefs fixed, Theorem 5 gives the posterior $q^\star(u; \gamma) = p(u)\, \nu_0(u)\, e^{\gamma \mathbb{I}_{t+1}(u)} / Z(\gamma)$ of the next input, where the message $\nu_0 > 0$ collects the expected goal log-densities and does not depend on γ, and $Z(\gamma)$ normalizes. This posterior is an exponential family with the natural parameter γ and the sufficient statistic $\mathbb{I}_{t+1}$ (Wainwright et al., 2008). The first derivative of its log-normalizer $\ln Z$ is therefore the mean $h(\gamma)$ of the information gain, and the second derivative is the variance $h'(\gamma)$. This variance is positive, because the information gain varies over sets of inputs with positive prior probability, which keep positive probability under the posterior. Hence h is continuous and strictly increasing. As γ grows, the factor $e^{\gamma \mathbb{I}_{t+1}}$ moves the posterior mass to the inputs whose information gain lies close to $\mathbb{I}^{\max}$, so $h(\gamma) \to \mathbb{I}^{\max}$, and strict monotonicity keeps $h(\gamma)$ below this limit.

The cases follow from the KKT conditions (15) with the constraint value (16). If $\beta_t \leq h(0)$, then $\gamma = 0$ meets them, and every $\gamma > 0$ violates complementary slackness, because $h(\gamma) > \beta_t$. Otherwise, $\gamma = 0$ violates the floor, so $\gamma > 0$ and complementary slackness requires $h(\gamma) = \beta_t$. For $\beta_t < \mathbb{I}^{\max}$, the intermediate value theorem and the monotonicity of h give exactly one root, whereas for $\beta_t \geq \mathbb{I}^{\max}$, no root exists, because $h < \mathbb{I}^{\max}$. Section S1.3 of the supplementary material gives the rigorous proof, including the bounds behind the derivatives and the limit. ■

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Generative artificial intelligence use

Generative artificial intelligence (AI) was used to assist with proving theorems, programming experiments and results figures. The output has been checked by the author, who takes responsibility for its correctness. AI assisted with initial drafting, but most text has been rewritten by the author.

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